

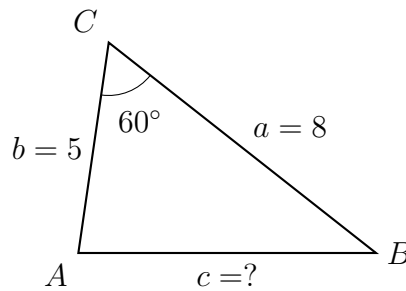
Solving SAS and SSS Triangles with the Law of Cosines

Finding a third side from SAS, an angle from SSS, a remaining angle after SAS, and distances in applied settings

Directions.

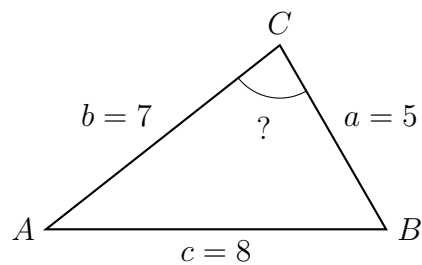
- Law of cosines: $c^2 = a^2 + b^2 - 2ab \cos C$, and the same shape with a or b in front. Side a is opposite angle A , and so on.
- Rearranged for an angle: $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$.
- Every triangle is drawn to scale from its own given values; the unknown is marked with a question mark.
- Round sides and angles to two decimal places. Keep full precision inside the calculation and round once, at the end.

1. Here $a = 8$, $b = 5$ and the included angle $C = 60^\circ$. Find c .



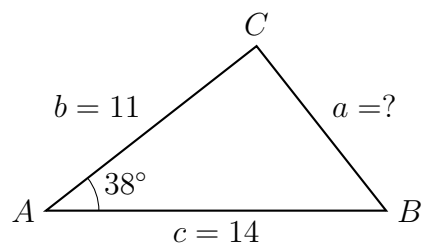
Answer: _____

2. Here $a = 5$, $b = 7$ and $c = 8$. Find angle C .



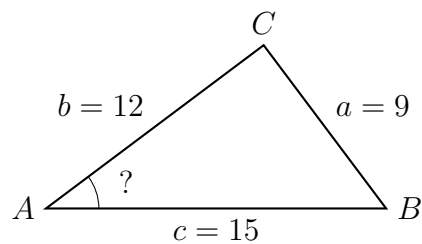
Answer: _____ degrees

3. Here $b = 11$, $c = 14$ and the included angle $A = 38^\circ$. Find a .



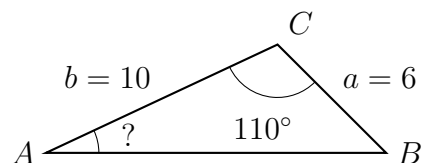
Answer: _____

4. Here $a = 9$, $b = 12$ and $c = 15$. Find angle A .



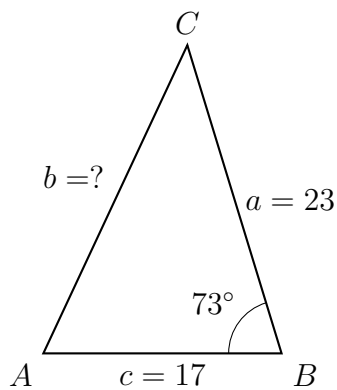
Answer: _____ degrees

5. Here $a = 6$, $b = 10$ and the included angle $C = 110^\circ$. Find angle A . (Find the third side first.)



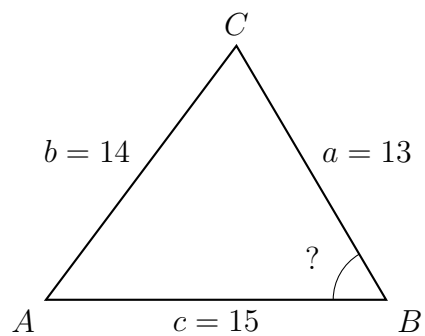
Answer: _____ degrees

6. Here $a = 23$, $c = 17$ and the included angle $B = 73^\circ$. Find b .



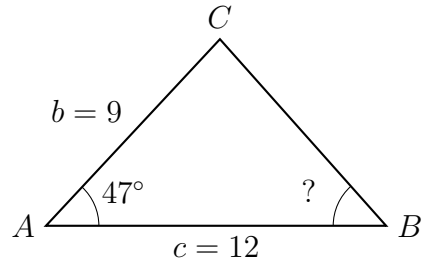
Answer: _____

7. Here $a = 13$, $b = 14$ and $c = 15$. Find angle B .



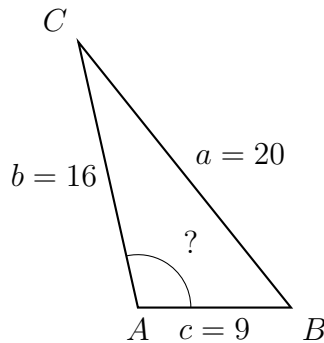
Answer: _____ degrees

8. Here $b = 9$, $c = 12$ and the included angle $A = 47^\circ$. Find angle B .



Answer: _____ degrees

9. Here $a = 20$, $b = 16$ and $c = 9$.



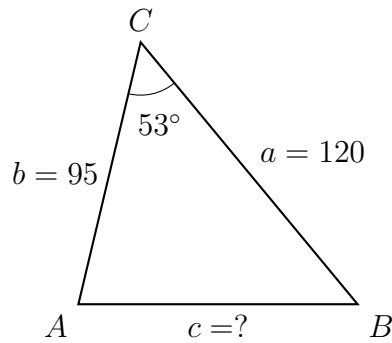
- (a) Find angle A .

Answer: _____ degrees

- (b) Before you took the inverse cosine, the value of $\cos A$ already told you something about angle A . Explain what, and why.

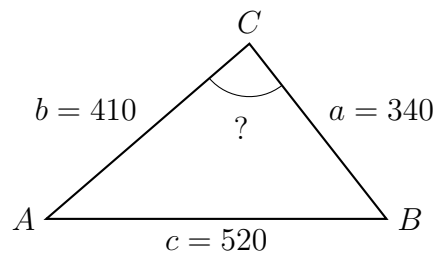
Answer: _____

10. Two paths leave a car park at an angle of $C = 53^\circ$ to one another. One runs $a = 120$ metres to a bird hide; the other runs $b = 95$ metres to a jetty. Find c , the straight-line distance from the hide to the jetty, in metres.



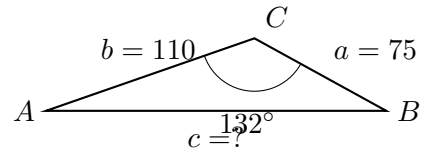
Answer: _____ m

11. Three survey markers form a triangle with side lengths $a = 340$ metres, $b = 410$ metres and $c = 520$ metres. Find the angle C at the marker opposite the longest side.



Answer: _____ degrees

12. From a launch pad a drone flies $a = 75$ metres out to a mast; on a second flight it leaves the same pad and flies $b = 110$ metres out to a tower. The angle at the launch pad between the two flight lines is $C = 132^\circ$.



- (a) Find c , the direct distance from the mast to the tower, in metres.

Answer: _____ m

- (b) Explain why the law of sines could not have been used as the first step here.

Answer: _____